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1 Evidence Fields: Visualizing Support, Contradiction, Confidence, and Provenance in AI Interfaces

> **Status: research draft (preprint, work in progress).** Paper 3 of the Fundamental family — a domain validator for evidence and trust in AI interfaces. Claims verified against the codebase and canonical docs as of 2026-06-07. See the series index and *the caveat canon* therein. This is a preprint draft, not canonical product documentation.

Author: Zach Shallbetter **Series:** Fundamental Research Papers, Paper 3 of 8 **Companion papers:** Fundamental: A Field Translation Runtime for Relational DOM Interfaces (flagship paradigm paper); Reading Field (reading attention and document memory). See the series index.

1.1 Abstract

AI assistants, retrieval systems, and other source-backed interfaces ask people to trust fluent prose. The dominant trust primitives they offer — *inline citations* and a *confidence badge* — are weak: a citation marks that *a* source was attached to *a* span, and a badge reports a scalar the reader has no way to calibrate, but

neither expresses the *relationships* that actually determine whether a claim should be believed. Which source supports which claim? Where do two sources contradict? Which sentence is unsupported entirely? How was this claim revised? These are relational facts, and current interfaces flatten them into footnote markers and a single number.

We argue that AI interfaces need a **relational evidence model**, and we show that Fundamental — the platform-native relational field runtime introduced in Paper 1 — can represent claims, sources, support, contradiction, uncertainty, provenance, and revision as *inspectable field relationships* rather than as decorations. In the **Evidence Field** model, claims are bodies; confidence is priority (gravity / strength); a source that supports a claim binds to it (strong-interaction binding, realized as a typed `supports` edge in the `RelationshipRegistry`); contradictory sources *repel* and raise the entropy of the claim they touch (electromagnetic opposition); uncertainty is the entropy metric; provenance is a memory trail; and a correction is a memory overwrite. Each mapping is classified by *truth mode*, and almost all of them are *semantic truth* — meaning mapped to a field metric — not physics.

We are deliberately more hedged than the flagship paper on two points. First, **Fundamental visualizes evidence; it does not adjudicate it.** The runtime surfaces the support, contradiction, and confidence that the *host* supplies; it has no fact-checking capability and makes no claim about whether a source is correct. Second, although the Evidence Field *pattern* and the data-binding mechanism (`bindData()`) that wires real claim/source records into the field both ship, and a data-bound Evidence Field *study page* exists as an interactive demonstration, **no controlled user-study results exist**. This paper therefore contributes a *model* and a *full study design*, clearly labeled as design, not measured findings. The study's centerpiece is **trust calibration** measured against ground truth (signal-detection d and a calibration error), with an explicit falsification path: a richer evidence display could *worsen* calibration by inducing over-trust, and the design is built to detect exactly that.

1.2 1. Introduction

1.2.1 1.1 The problem: trust primitives in AI interfaces are weak

A modern AI answer is a paragraph of fluent text, occasionally annotated with superscript citation markers and, sometimes, a confidence indicator. These two devices carry almost all of the interface's burden of trust, and both are easy to ignore and structurally impoverished:

- **Inline citations** assert that a source was *attached to a span*. They do not say whether the source *supports* the span, *partially* supports it, is *about* it but neutral, or in fact *contradicts* a different span nearby. Citation markers are also easy to skip: a reader who trusts the prose has no reason to open them, and the marker looks identical whether the underlying source is a peer-reviewed study or an unrelated blog post.
- **Confidence badges** report a scalar — "high confidence," 0.82, three of five bars — that the reader cannot calibrate. A badge says nothing about *which* claims in the answer are confident and which are not, and a single number for a multi-claim answer is a category error: it averages over claims that may individually range from well-established to fabricated.

Crucially, neither primitive expresses *relationships*. The facts that matter for trust are relational: *this* source supports *that* claim; *these two* sources contradict each other; *this* sentence is unsupported by anything; *this* answer was revised after a correction. Inline citations and badges encode none of this. The reader is left to reconstruct the evidence graph by hand, from flat text, under exactly the conditions — speed, fluency, plausibility — that discourage them from doing so.

1.2.2 1.2 Automation bias and over-trust of fluent output

The cost of weak trust primitives is not neutral. A large literature on **automation bias** and over-reliance on automated decision aids documents that people tend to over-trust confident, fluent system output and under-scrutinize it, particularly under time pressure [parasuraman1997; skitka1999; goddard2012]. Large

language models compound the risk: they produce fluent, well-formed prose *independently* of whether the underlying claims are supported, and studies of citation faithfulness in retrieval-augmented systems report that attached citations frequently do not entail the claims they decorate [liu2023verifiability; gao2023alce; wallat2024faithfulness]. The interface, then, is being asked to communicate the *one thing the text itself cannot convey*: where the prose is and is not anchored in evidence — and it does so with a footnote marker and a bar.

This is the gap the Evidence Field targets. The claim is narrow and we keep it narrow throughout: AI and source-backed interfaces need a *relational* evidence model, and Fundamental can provide one in which support, contradiction, confidence, and provenance are first-class, inspectable field relationships rather than inline ornaments.

1.2.3 1.3 The Evidence Field idea

Fundamental's organizing principle (Paper 1, §3) is reciprocal: *elements bend the field; the field bends them back.* Applied to evidence, the idea is this. Treat each **claim** in an answer as a *body.* Treat each **source** as a body. Treat **support** as a binding relationship between a source and a claim, and **contradiction** as an opposing one. Let the field then write back, onto each claim, a continuous measure of how *coherent* (well-resolved, well-supported) or how *uncertain* (conflicted, unsupported) it is — surfaced as CSS custom properties and thresholded events that drive the claim's visual weight. An unsupported claim, bound to nothing, drifts and reads as unstable; a claim ringed by mutually supporting sources reads as coherent; two sources that contradict push apart and raise the local entropy of the claim between them. The evidence graph that a reader would otherwise have to reconstruct from footnotes becomes a visible, inspectable field.

The decisive design constraint, repeated throughout this paper, is that **the field visualizes evidence the host supplies; it does not establish truth.** The system has no oracle. Whether a source actually supports a claim, and whether the claim is actually correct, are judgments the host application (or its model, or its human authors) must make and feed in. Fundamental's contribution is to make those supplied judgments *relational, legible, inspectable, and accessible* — not to make them.

1.2.4 1.4 Contributions

This paper contributes:

1. **A relational evidence model** (§3): a mapping from evidence semantics — claim, source, support, contradiction, confidence, uncertainty, provenance, revision — onto Fundamental bodies, relationships, and metrics, grounded in the shipped Evidence Field pattern family and classified by truth mode. The model's honesty constraint (visualize, do not adjudicate) is made concrete by the code: the platform *derives* coherence and entropy from relationship resolution, while a claim's *confidence* is carried from the host — since #220 the engine computes **no** confidence fallback, so confidence is present only when supplied, and is never adjudicated.
2. **A full controlled user-study design** (§5): research questions, hypotheses, three conditions (a current-practice baseline, the Evidence Field, and a reduced-motion variant), a curated ground-truth corpus, tasks and measures keyed to the project's stated evaluation targets, a primary trust-calibration outcome (signal-detection d plus a calibration error), a power estimate, an analysis plan, threats to validity, and a pre-registration commitment.
3. **An explicit falsification path** (§6): the prediction that a richer evidence display might *worsen* calibration through over-trust, and the specific way the design would detect it.

We defer the runtime architecture to Paper 5, the data-binding mechanics to Paper 7, the reduced-motion conformance model to Paper 4, and diagnostics depth to Paper 8, cross-referencing rather than re-explaining.

1.3 2. Background and related work

The Evidence Field sits at the intersection of several lines of work. We position it against each; keys below correspond to verified entries in `references.md`; no citation is invented solely to satisfy the paper.

Trust calibration in decision support. Work on human–automation trust holds that the goal is not maximal trust but *calibrated* trust — reliance that tracks the system's actual reliability — with miscalibration taking two forms, over-trust and under-trust [lee2004]. This frames our primary dependent variable: an evidence display is judged by whether it moves users toward calibration, not by whether it raises confidence.

Uncertainty visualization. Visualizing uncertainty is hard: representations that *look* authoritative can suppress appropriate doubt, and the encoding choice changes how much uncertainty people perceive and act on [hullman2019; kale2019]. The Evidence Field encodes uncertainty as a continuous entropy metric written back to the claim — one such representation, and it inherits these risks; §6 treats them as a falsification path, not a guarantee.

Automation bias and over-reliance on AI. People over-rely on confident automated output, especially under load [parasuraman1997; skitka1999; goddard2012]; recent work on over-reliance on LLM assistants asks whether explanations and confidence signals reduce or *increase* unwarranted reliance, with mixed results [bansal2021; bucinca2021; kim2025llm]. We take the mixed results seriously: a relational display is not assumed to help.

Source credibility and citation behavior. Studies of credibility assessment and citation inspection suggest attached references are frequently not opened and that credibility cues are often heuristic rather than evidential [hovland1951; fogg2003; jurgens2016]. The Evidence Field's premise is that making support *structural and visible*, rather than a marker the reader must choose to open, could change this; the study tests it.

LLM hallucination and citation faithfulness. Retrieval-augmented generation attaches sources, but evaluations report that a model's cited source frequently does not *entail* the sentence it is attached to [liu2023verifiability; gao2023alce; wallat2024faithfulness] — precisely the relationship the Evidence Field externalizes: a "supports" edge can be *present but weak*, and a contradiction can sit unmarked between two attached sources.

Explanation usefulness and explainable AI. Work on whether explanations *help* (vs. merely satisfy) cautions that explanation can produce *illusory* understanding and that a plausible rationale can raise trust without improving decisions [bansal2021; bucinca2021; abdul2018]. We treat "explanation usefulness" as a measured construct in §5, not an assumed benefit.

Position against the baseline. Against the badge-and-citation baseline, Fundamental's stance is *relational and inspectable*: evidence is a typed graph the reader can see and probe, not a per-span marker plus a scalar. This inherits the flagship system's epistemic posture (truth modes, passports, conformance; Paper 1, §6) and the explainability framing developed in Paper 8 — every visible evidence behavior is traceable to a field cause. Whether the relational display measurably improves *calibrated* trust over the baseline is the open empirical question this paper is designed to answer, not to assert.

1.4 3. The Evidence Field model

1.4.1 3.1 The honesty constraint, stated first

Because every paragraph that follows discusses trust, we lead with the disclaimer and then never relax it:

> **Fundamental visualizes evidence; it does not adjudicate it.** The runtime surfaces support, contradiction, confidence, and provenance *supplied by the host*. It performs no fact-checking, verifies no source, and asserts no claim's correctness. "Evidence" here names a set of inspectable relationships, not an automated judgment of truth.

This is not only rhetoric; it is enforced by the code. The platform metric library (`packages/dom/src/metrics.ts`) is explicit about provenance: `coherence` and `entropy` are *derived* from how relationships resolve (the ratio of resolved to total edges, minus the conflicting ratio), but `confidence` is **supplied** by the host through `data-field-<metric>` and — since #220 — has **no computed fallback at all**: the engine never infers confidence from relationship presence (a citation is not certainty), so `--field-confidence` is written only when the host supplies it. (only `priority` retains a computed default; `risk`, like `confidence`, is supplied-only and intentionally has no default — see `SUPPLIED_ONLY_METRICS` in the metric library.) A claim's confidence enters the field because the application put it there. The field's job is to make the *consequences* of that supplied evidence — which claims read coherent, which read contested — visible and consistent.

1.4.2 3.2 The Natural Field Translation, applied to evidence

Paper 1 (§6.5) describes the Natural Field Translation System: Fundamental does not copy physics into the interface, it *translates* the four fundamental fields into interface grammar — gravity → priority, electromagnetism → polarity and signal, strong interaction → binding, weak interaction → transformation. Evidence is a near-ideal domain for this translation, because an evidence graph *already has* priority (confidence), polarity (support vs. contradiction), binding (a source backing a claim), and transformation (revision and decay). The canonical natural-fields document already files the Evidence Field under *electromagnetic + strong* (`docs/canonical/natural-fields.md`), and the shipped pattern agrees.

The mapping, with each row's truth mode:

Evidence semantics	Field representation	Engine vocabulary
Claim	a body	`[data-body]` element
Confidence (of a claim)	priority / strength	gravity translation; `--field`-weight via str
Source	a body	`[data-body]` element
Support (source → claim)	strong-interaction **binding**	`link`, `cohesion`; a typed `supports` edge
Contradiction (source source / claim)	electromagnetic **opposition**	`charge` + `repel`; raises entropy
Uncertainty	the **entropy** metric	`--field-entropy` (derived)
Unsupported claim	unstable, bound to nothing → drift	low coherence; high entropy
Provenance	a **memory trail** / path	`memory`; Citation Thread / Provenance Tra
Correction / revision	**memory overwrite**	the memory-deposit update rule
Verified claim	**coherence**	`--field-coherence` (derived)
Active generation	stream / `fieldflow`	transport along structure

Almost every row is *semantic truth* (meaning → metric), by design. This matters for honesty: the Evidence Field is not claiming to be physics. A "supports" edge does not obey a conservation law; it is a translation of an editorial judgment into a field relationship so the field can render it consistently. The one place a stricter register appears is contradiction, which borrows the electromagnetic *polarity* idea — opposing charges push apart — but even there the discipline of Paper 1's electromagnetic rule holds: **electric pushes, magnetic bends, fieldflow carries.** Contradiction uses `charge`/`repel` to *push* conflicting bodies apart and to raise entropy; it does *not* misuse `magnetism` (which bends moving charge and does no work) or `fieldflow` (transport along structure). Keeping these lanes separate is what lets the model stay legible.

1.4.3 3.3 The shipped Evidence Field pattern

The Evidence Field is pattern #5 in the 64-recipe gallery and ships as a validated `FieldPattern` (`packages/core/src/recipes/catalog.ts`). Its definition, verbatim in substance:

```
export const EVIDENCE_FIELD: FieldPattern = {
  id: 'evidence-field',
  name: 'Evidence Field',
  intent: 'show how sources support, weaken, or contradict a claim',
```

```
  naturalField: 'electromagnetic',           // electromagnetic (+ strong)
  primitives: ['charge', 'link', 'cohesion', 'repel'],
  bodies: [
    { body: 'charge',   strength: 0.9, range: 280, spin: 1 },
    { body: 'link',     strength: 0.7, range: 320 },
    { body: 'cohesion', strength: 0.6, range: 260, feedback: true },
    { body: 'repel',    strength: 0.5, range: 200 },
  ],
  relationships: [{ from: 'claim', to: 'source', type: 'supports', strength: 0.7 }],
  render: ['links', 'particles', 'heatmap'],
  metrics: ['coherence', 'entropy'],
  diagnostics: ['topology', 'causality', 'links'],
  accessibility: {
    reducedMotion: 'a static claim/source table with support and conflict badges',
    meaningWithoutMotion: 'each source is listed as supporting or contradicting, with a confidence label
  },
  notes: 'Claims are bodies; supporting sources bind them (link + cohesion), contradictory sources ' +
         'repel and raise entropy (electromagnetic + strong). Strong evidence increases coherence.',
};
```

Every token here is real and passported (the pattern conformance gate, Paper 1 §7.3, rejects any pattern whose primitives are not real passported tokens, whose render layers or diagnostics are not real modes, or whose declared primitives drift from its body tokens). The four bodies translate the evidence semantics directly: `charge` sets the support/contradiction polarity; `link` and `cohesion` *bind* a supporting source to its claim (the strong-interaction translation); `repel` pushes a contradicting source away and is what raises local entropy. The pattern's two declared `metrics` — `coherence` and `entropy` — are exactly the two the platform *derives* from relationship resolution, so the pattern's output is grounded in supplied edges, not invented. The `feedback: true` flag on `cohesion` opts the binding into the reverse write-back (Paper 1, §8).

The Evidence Field does not stand alone. Five sibling patterns in the same family complete the evidence vocabulary, all shipped and all validated:

Pattern (shipped)	Natural field	What it shows	Primitives
Evidence Field	electromagnetic (+ strong)	support, contradiction, claim coherence	`charge`, `lin`
Trust Gradient	electromagnetic	confidence, verification, unsupported claims	`charge`, `lin`
Source Constellation	strong	sources clustering around claims/topics	`link`, `charg`
Citation Thread	electromagnetic	citations/footnotes/references as visible relationships	`link`, `field`
Provenance Trail	strong	origin + transformation history of content	`memory`, `lin`
Conflict Field	weak (+ electromagnetic)	contradiction, uncertainty, unstable state	`charge`, `rep`

Read together they belong to the canonical *Signal Path* theme — "citations, dependencies, evidence" (`natural-fields.md`) — though at the token level the evidence patterns draw on `charge`, `link`, `cohesion`, `repel`, `memory`, `morph`, and `diffuse` rather than the Signal Path triad (`charge`/`propagate`/`fieldflow`) itself. Three are worth a line. **Trust Gradient** is where confidence and the *unsupported-claim* surface live — its metrics are `trust`, `confidence`, `coherence`, `entropy`, and its required reduced-motion form is "trust badges with an evidence table and an unsupported-claim list." **Provenance Trail** carries `memory` at strength 1.1 (the family's highest) because provenance *is* a persistence metric — the memory trail of §3.2 — with `morph` carrying the transformation history (the weak-interaction translation). **Conflict Field** isolates contradiction: opposing states `repel` and raise entropy, while `morph` and `diffuse` carry the transformation pressure toward resolution or a decayed warning.

1.4.4 3.4 What writes back to the DOM

The reverse half of the loop — field → element — is where the model becomes legible to a reader (Paper 1, §8). For evidence, the load-bearing write-backs are two derived scalars, `--field-coherence` and `--field-entropy`, both of which are real platform output variables (`packages/core/src/visual/tokens.ts`, `FIELD_OUTPUT_VARS`). The platform computes them per frame from each claim's relationship neighborhood (`computeMetrics` in `packages/dom/src/metrics.ts`):

```
resolvedRatio = relResolved / relTotal
conflictRatio = relConflict / relTotal
coherence     = clamp01(resolvedRatio − conflictRatio)
entropy       = clamp01(conflictRatio + (1 − resolvedRatio) · 0.5)
```

Since #222, this resolution is **real**: `applyRecipe` counts `relResolved` as the edges whose endpoints actually resolve and `relTotal` as resolved + declared-but-unresolved, so a citation pointing at nothing lowers `resolvedRatio`, lowers coherence, and *raises* entropy (the `(1 − resolvedRatio)` term is live, not dead code). The registry now tracks the unresolved declarations rather than silently dropping them, and inspection can name each missing endpoint.

A claim with many resolved supporting edges and no contradictions reads as coherent (high `--field-coherence`, low `--field-entropy`); a claim touched by contradictions reads as contested; a claim with no edges at all stays low-coherence and inherits its prior entropy — the *unsupported* case. Authors map these scalars to type weight, color, and a stability cue in pure CSS, exactly as the flagship's "material typography" does for density. Crucially — and this is the honesty constraint again, now mechanical — `confidence` is *not* in that derivation: it is supplied by the host (`data-field-confidence`) and merely *carried*, never manufactured. The field can make a supplied low-confidence claim *look* unstable; it cannot decide that the claim is false.

Thresholded events complete the write-back: a claim crossing into high entropy can emit an `entropy-warning`-class event (the canonical event vocabulary includes `field:entropy-warning`), and a binding strengthening can emit a relationship event. These are debounced and inspectable, per the interaction model. The relationship graph itself is owned by the `RelationshipRegistry` (Paper 1, §5.2; `docs/canonical/interaction-and-relationship-model.md`, §7), which resolves a `supports` or `contradicts` edge from native DOM signals (`href`, ARIA references, `data-field-relation`) into a typed graph mapped onto core `RelationshipAgent`s, with a `relation-target-missing` lint rule flagging any edge whose target is not a registered body.

1.5 4. Implementation status

A preprint earns trust by being precise about what is real. The verify-against-code rule (README) governs here, and we apply it strictly — including where the codebase has moved *ahead* of earlier planning notes.

Shipped (verifiable in the pattern catalog, the platform exports, and the tests as of 2026-06-07):

- The **Evidence Field pattern** and its five siblings (Trust Gradient, Source Constellation, Citation Thread, Provenance Trail, Conflict Field) ship as validated `FieldPattern`s in `packages/core/src/recipes/catalog.t` gated by `validateRecipe` (`packages/core/src/recipes/schema.ts`).
- The **coherence/entropy metric derivation** and the supplied-vs-derived metric discipline ship in `packages/dom/src/metrics.ts`, with `--field-coherence` / `--field-entropy` as real output variables.
- **`bindData()`** — the data-binding mechanism that wires real claim/source records into the field — ships in `@fundamental-engine/dom` (`packages/dom/src/bind-data.ts`, with `apply-recipe.ts` and a test suite). Records become bodies, mapped metrics become state, and mapped relationships become graph edges, with deterministic id-diffed updates and decay-on-remove. (Its mechanics are the subject of Paper 7; we use it here only as the evidence wiring.)
- A **data-bound Evidence Field study page** exists as an interactive demonstration at `apps/site/src/pages/docs/studies/evidence-field.astro`: claims are records, each map-

ping to a body with `supports`/`contradicts` relationships to its sources; the `evidence-field` pattern turns those edges into coherence/entropy; flipping a source's polarity or adding/removing a claim updates the field deterministically; and a reduced-motion path is wired in.

Not shipped, and the reason this paper is a design: no controlled user-study results exist. The study page is a *demonstration of the mechanism*, not an *experiment*. There is no curated ground-truth corpus, no participant pool, no condition assignment, no measured trust calibration — none of the apparatus §5 specifies. Caveat-canon item 6 holds in full: every empirical claim in this paper is a *hypothesis* or a *design*, never a finding.

We flag a discrepancy honestly for the series lead. The roadmap archive (`docs/planning-archive/roadmap-frontiers.md`, where data binding is frontier **R5** and AI evidence fields are **R14**) framed `bindData()` and the Evidence Field study as *unbuilt*. The code has since advanced: `bindData()` and the data-bound study page both exist (commits #210 and #213 in the repository history). The verify-against-code rule means the code wins, so this paper reports the *mechanism and the demo as shipped* while holding firmly that the *empirical study and its results do not exist*. The contribution remains a model plus a study protocol; the demonstration strengthens the model's plausibility but is not evidence of user benefit.

1.6 5. A user-study design

This is the paper's centerpiece. We design a full controlled study to test whether the Evidence Field improves *calibrated* trust in AI answers relative to current practice. It is a **design and a protocol**; it has not been run.

1.6.1 5.1 Research questions and hypotheses

The motivating question, in the project's own words, is *"Does Evidence Field improve trust calibration?"* (`docs/canonical/field-possibilities.md`). We refine it into five questions, three of which map to an evaluation target the project already names (source-support recognition, perceived distraction, accessibility preference), and two of which — contradiction detection and unsupported-claim identification — extend that set:

- **RQ1 (support recognition).** Does the Evidence Field improve readers' accuracy at identifying *which source supports which claim*?
- **RQ2 (contradiction detection).** Does it improve detection of *contradictions* between sources?
- **RQ3 (unsupported-claim identification).** Does it improve identification of claims that are *unsupported by any source*?
- **RQ4 (trust calibration — primary).** Does it move readers' trust toward *calibration* with ground truth — i.e. raise discrimination between correct and incorrect claims (d) and reduce calibration error — relative to the baseline?
- **RQ5 (confidence interpretation & explanation usefulness).** Does it improve readers' interpretation of supplied confidence, and do they rate the evidence display as *useful* for deciding whether to rely on the answer?

Directional hypotheses (predictions, restated as falsifiable in §6):

- **H1–H3.** Evidence Field > Baseline on support recognition, contradiction detection, and unsupported-claim identification.
- **H4 (primary).** Evidence Field improves calibration over Baseline: higher d, lower calibration error (Brier score / expected calibration error, ECE).
- **H4 (the danger hypothesis, pre-registered as a two-sided possibility).** The Evidence Field may *worsen* calibration via over-trust — a richer, authoritative-looking display could raise reliance on *incorrect* claims. We pre-register over-trust rate as a primary safety outcome, not a footnote.
- **H5.** The reduced-motion Evidence Field is *non-inferior* to the full Evidence Field on RQ1–RQ4 (meaning survives without motion; Paper 4).

1.6.2 5.2 Conditions

A between-subjects (or counterbalanced within-subjects; see §5.6) comparison of three displays of the *same* answers:

1. **Baseline — current practice.** The AI answer as prose with **inline citation markers and a confidence badge**. This is the honest state of the art, not a strawman: real citations the reader can open, and a per-answer confidence indicator.
2. **Evidence Field.** The same answer, same sources, rendered with the Evidence Field pattern family via `bindData()`: claims are bodies, support/contradiction are visible typed edges, coherence and entropy are written back per claim, and unsupported claims read unstable. Confidence is the *same supplied* value as the baseline — the only thing that changes is that evidence is *relational and visible* rather than a marker plus a number.
3. **Evidence Field, reduced motion.** The pattern's required static equivalent: the claim/source table with support and conflict badges, the unsupported-claim list, and confidence labels — no travel. This condition controls for the possibility that any benefit is a motion/novelty artifact and ties the study to the accessibility conformance claim of Paper 4.

All three render identical *content* and identical *supplied confidence*; they differ only in how evidence relationships are expressed. This isolates the relational-display manipulation.

1.6.3 5.3 The corpus: AI answers with ground-truth annotations

The study requires a **curated corpus** of AI answers for which the evidence relationships are known, because the dependent variables are scored against ground truth. Each answer is decomposed and annotated by independent expert annotators (with inter-annotator agreement reported) along:

- **Claims** — the answer segmented into atomic, individually-verifiable claims.
- **Sources** — the source set attached to (or available for) the answer.
- **Support / contradiction labels** — for each (claim, source) pair: *supports*, *contradicts*, or *neutral/irrelevant*, with a strength.
- **Ground-truth claim status** — each claim labeled **correct**, **incorrect/hallucinated**, or **unsupported** (no source in the set entails it), established independently of the model's own citations.

The corpus is **balanced**: every answer contains a mix of well-supported-correct, contradicted, and unsupported/hallucinated claims, so the tasks cannot be gamed by a constant response. Item difficulty is piloted and items are stratified across difficulty. Domain coverage spans at least three domains (e.g. a research-summary domain, a legal/policy-claim domain, and a product-support domain) so that effects are not an artifact of one topic. The corpus, its annotation guidelines, and agreement statistics are released with the pre-registration.

1.6.4 5.4 Tasks and measures

Tasks map directly to the project's evaluation targets (*source-support recognition*, *perceived distraction*, *accessibility preference*; `field-possibilities.md` §20) and to the RQs:

Task	Measures
Match each claim to its supporting source(s)	**source-support recognition** accuracy (and d over (claim, sou
Flag pairs of sources that contradict	**contradiction detection** — hit rate, false-alarm rate, d
Mark claims unsupported by any source	**unsupported-claim identification** — hit/false-alarm, d
Decide, per claim, whether to rely on it; rate confidence	**trust calibration** — d (correct vs. incorrect), **Brier score /**
Interpret the supplied confidence; rate display usefulness	**confidence interpretation** accuracy; **explanation-usefulnes**

Primary DV — trust calibration. For each participant we collect, per claim, a reliance decision and a subjective confidence. From these and the ground-truth claim status we compute:

- **Discrimination, *d*** (signal-detection): treating "claim is correct" as signal and the reliance decision as the response, d measures the reader's ability to *separate* correct from incorrect claims. Higher d = better discrimination. We report criterion c alongside, because a display can shift *bias* (toward trusting everything) without improving discrimination — exactly the over-trust signature.
- **Calibration error** — the **Brier score** of probabilistic confidence against the binary correctness outcome, and **expected calibration error (ECE)** from a reliability binning. Lower = better-calibrated. We report a reliability diagram per condition.

Secondary measures. Over-trust rate (reliance on incorrect/unsupported claims) and under-trust rate (rejection of correct, well-supported claims); decision time per claim; subjective workload (a standard multidimensional workload instrument) [hart1988]; perceived distraction; and an accessibility-preference item for the reduced-motion condition. Process data — whether and when the reader opens a citation in baseline, or probes an edge / opens the inspector in the Evidence Field — is logged to interpret the outcome measures.

1.6.5 5.5 Power and sample-size estimate (planning estimate)

A planning estimate only; the registered analysis fixes the final numbers. Treating the primary contrast (Evidence Field vs. Baseline on calibration) as a between-groups comparison and targeting a *medium* standardized effect (Cohen's d 0.5) at = 0.05, two-sided, with 80% power, a standard two-group calculation gives roughly **64 participants per condition** (128 for the two-condition core contrast; 192 across all three conditions). A within-subjects design with counterbalancing (§5.6) would reduce this materially by removing between-person variance, at the cost of carryover risk. Because the *danger* hypothesis (H4 , over-trust) is a primary safety outcome and may be a smaller effect, we will power for the *smaller* of the two primary effects and treat the per-condition n as a floor, not a target. The number is a planning figure, not a result.

1.6.6 5.6 Procedure

After consent and a calibration warm-up (unrelated items, to familiarize participants with the reliance-and-confidence response format), each participant works through a stratified sample of corpus answers in their assigned condition, performing the §5.4 tasks per answer. A within-subjects variant counterbalances condition order across answers (Latin square) and inserts a washout, to gain power while guarding against carryover; the between-subjects variant avoids carryover entirely at the cost of n. The reduced-motion condition is run under an enforced `prefers-reduced-motion` setting. A short post-task interview (qualitative, on a subsample) probes *why* participants trusted or distrusted specific claims, to interpret the calibration numbers. The full protocol, instruments, and the corpus are deposited with the pre-registration before any data is collected.

1.6.7 5.7 Analysis plan

Confirmatory analyses are specified in advance and limited:

- **Primary.** Mixed-effects models with participant and item as crossed random effects: d and calibration error (Brier/ECE) as outcomes, condition as the fixed effect, planned contrasts Evidence Field − Baseline (H4) and Evidence Field − reduced-motion (H5, an equivalence/non- inferiority test against a pre-registered margin).
- **Safety.** Over-trust rate as a primary outcome with the *same* model and an explicit pre- registered interpretation: a *significant increase* in over-trust under the Evidence Field, even with improved discrimination, is recorded as a partial failure (§6).
- **Secondary/exploratory.** Support recognition, contradiction detection, unsupported-claim identification, decision time, workload, distraction, accessibility preference; clearly labeled exploratory, with multiplicity control. Process-data analyses (citation opens, edge probes, inspector use) are exploratory mediators.

Effect sizes with confidence intervals are reported for every contrast; we do not treat $p < .05$ as the finding. A reliability diagram per condition accompanies the calibration numbers.

1.6.8 5.8 Threats to validity

- **Automation bias / demand.** A novel, attractive display may induce participants to *perform* scrutiny or, worse, to over-rely; the reduced-motion condition, the over-trust safety outcome, and process logging are the guards.
- **Novelty effect.** First exposure to a field display could inflate (or depress) engagement; a warm-up, a within-subjects washout, and reporting of order effects address it.
- **Corpus bias.** Results may not generalize beyond the curated corpus; multi-domain coverage, difficulty stratification, released annotation guidelines, and reported inter-annotator agreement mitigate but do not eliminate this. The corpus's *construction* could itself favor the relational display (e.g. if contradictions are made unusually salient); annotation is therefore independent of the display design.
- **The richer-display-induces-over-trust risk — addressed directly.** This is the study's central threat and is elevated to a hypothesis (H4) and a primary outcome, not buried. A display that makes evidence *look* thorough could raise reliance on *incorrect* claims. The design detects it through the d /criterion decomposition (a bias shift without a discrimination gain is the over-trust signature) and through the explicit over-trust rate. If the Evidence Field raises confidence and reliance without raising discrimination, the study reports that as evidence *against* the system, not for it.
- **Construct validity of "calibration."** Reliance and confidence are imperfect proxies for trust; we triangulate with the qualitative interview and report both behavioral (reliance) and subjective (confidence) calibration separately.

1.6.9 5.9 Pre-registration

The hypotheses (including H4), conditions, corpus, primary and secondary measures, the d and Brier/ECE operationalizations, the power floor, the analysis models, and the stopping rule are **pre-registered** before data collection, on a public registry, with the corpus and analysis code deposited. Deviations are reported as deviations. This is stated as a commitment of the design, not as something already done.

1.7 6. Hypothesized outcomes

These are *predictions*. No data exists; nothing here is a finding.

If the model's premise holds, we would expect the Evidence Field to improve discrimination (higher d) and reduce calibration error (lower Brier/ECE) relative to the badge-and-citation baseline, with the largest gains on **unsupported-claim identification** (the case the baseline expresses worst — an unsupported claim looks identical to a supported one in plain prose) and on **contradiction detection** (which the baseline does not express at all). We would expect the reduced-motion condition to be non-inferior, supporting the Paper 4 claim that meaning is not motion. We would expect process data to show readers *probing* the evidence graph (opening edges, the inspector) more than they *open citations* in the baseline, and the explanation-usefulness rating to favor the Evidence Field.

The falsification case, stated as plainly as the success case. The Evidence Field could *worsen* calibration. The mechanism is over-trust: a relational display that *looks* authoritative and thorough may raise readers' reliance on the answer wholesale — including on the incorrect and unsupported claims it contains — even as it makes the supported ones easier to see. The signal-detection decomposition is designed precisely to catch this: **over-trust appears as a shift in criterion c (a bias toward reliance) without a corresponding gain in d (discrimination).** If the data show the Evidence Field raising confidence and reliance while d is flat or lower and the over-trust rate climbs, the honest conclusion is that the richer display *harmed* calibration, and the paper would report that the relational evidence model, *as rendered*, fails its central claim. A second failure mode is *no effect* — readers ignore the field as they ignore citations — which the design also detects (null d difference, unchanged over-trust). We commit in advance to reporting either outcome. An inspectable trust display that quietly increases over-trust would be worse than the baseline, and the study is built to say so.

1.8 7. Limitations

We restate the relevant caveat-canon items (README) and the model's specific limits.

1. **Visualization is not verification.** The Evidence Field surfaces the support, contradiction, and confidence *the host supplies*; it adjudicates nothing. A host that supplies wrong support labels will produce a confident, coherent-looking field around a false claim. The system's value is conditional on the quality of the supplied evidence, and the study's corpus is annotated *independently of the display* precisely so the evaluation is not circular.
2. **No user-study results exist** (caveat-canon item 6). The pattern family ships, `bindData()` ships, and a data-bound demonstration page ships — but the controlled study of §5 has not been run. Every empirical statement in this paper is a hypothesis or a design.
3. **The danger is real and unresolved.** The over-trust failure mode (§6) is not a hypothetical we can dismiss; the uncertainty-visualization and over-reliance literatures (§2) give concrete reason to expect that a richer display *can* hurt. Until the study is run, the model's benefit is genuinely open.
4. **Generalization across domains is unestablished.** Evidence semantics differ across research, legal, support, and journalistic contexts; the model is domain-agnostic by design, but its usefulness in any one domain is an empirical question the multi-domain corpus only begins to address.
5. **Semantic, not physical, truth** (caveat-canon item 4). The evidence mappings are *semantic* translations (meaning → metric), not physical laws; a "supports" edge is a rendered editorial judgment, and coherence/entropy are designed metrics, not conserved quantities. We keep the register explicit rather than dressing the model as physics.
6. **Scope.** Runtime architecture (Paper 5), data-binding mechanics (Paper 7), reduced-motion conformance (Paper 4), and diagnostics depth (Paper 8) are deferred; this paper validates one claim — that a relational evidence model is expressible and worth testing — and nothing more.

1.9 8. Discussion

The reason this matters is narrow and, we think, urgent: **the trust primitives shipping in AI interfaces today are weak**, and they are weak in a *structural* way that more polish will not fix. A confidence badge cannot express *which* claim is uncertain; an inline citation cannot express *whether* the source supports the span or *contradicts* a neighbor; neither can show that a sentence is supported by *nothing*. Those are relational facts, and a relational substrate is the natural place to put them. Fundamental already has that substrate — typed relationships in the `RelationshipRegistry`, derived coherence/entropy metrics, a reverse write-back, and a conformance discipline that keeps the vocabulary honest — so the Evidence Field is less a new system than a *reading* of an existing one onto the domain where weak trust primitives hurt most.

We are deliberately modest about what this buys. The Evidence Field is a *display*, and the central risk of any trust display is that it manufactures confidence rather than calibrating it. We have therefore refused to claim a benefit, designed the study so its primary outcome is *calibration* (not confidence), elevated the over-trust failure mode to a primary safety outcome, and built the signal-detection decomposition specifically to expose a display that increases reliance without increasing discrimination. The honest summary is that Fundamental makes a relational evidence model *expressible, inspectable, and accessible*, and that whether that model helps real readers make better-calibrated trust judgments is an open empirical question with a real chance of a negative answer — which is exactly why the study is worth running.

The discipline that makes the model trustworthy *as software* — the pattern conformance gate, the truth-mode classification, the supplied-vs-derived metric boundary, the required reduced-motion equivalent — is the same discipline that makes it trustworthy *as a research object*: every claim about what the Evidence Field does is checkable against the catalog, the metric library, and the relationship registry, not against prose.

1.10 9. Conclusion

AI and source-backed interfaces communicate trust through inline citations and confidence badges that are easy to ignore and cannot express the relationships that actually govern belief: which source supports which claim, where claims contradict, what is unsupported, how confident, and what the provenance is. We presented the **Evidence Field**, a relational evidence model in which claims and sources are bodies, support is binding, contradiction is electromagnetic opposition, uncertainty is entropy, provenance is a memory trail, and revision is a memory overwrite — each classified by truth mode, almost all of them semantic, and all of them *visualized, never adjudicated*. The Evidence Field pattern and its five siblings ship and are conformance-validated; `bindData()` and a data-bound demonstration page ship; **no controlled user-study results exist**. The contribution is therefore a model and a full, pre-registerable study design whose primary outcome is *trust calibration* against ground truth — and which is built, above all, to detect the failure case in which a richer evidence display makes trust *worse* by inducing over-trust. Whether the model helps is the question; the paper's job is to make that question answerable, honestly.

1.11 Appendix A. Reproducibility

Every model claim in this paper is checkable against the repository.

- **The Evidence Field pattern family** (Evidence Field, Trust Gradient, Source Constellation, Citation Thread, Provenance Trail, Conflict Field): `packages/core/src/recipes/catalog.ts`.
- **Pattern schema and conformance gate** (`validateRecipe`, the strict primitives/render/diagnostic checks, the required accessibility equivalent): `packages/core/src/recipes/schema.ts`.
- **The metric library** — the supplied-vs-derived boundary, and the coherence/entropy derivation that grounds "visualize, don't adjudicate": `packages/dom/src/metrics.ts`.
- **Output variables** (`--field-coherence`, `--field-entropy` in `FIELD_OUTPUT_VARS`): `packages/core/src/visual/tok`
- **The relationship graph** (`supports` / `contradicts` edges, native-signal resolution, the `relation-target-missing` lint): `packages/dom/src/relationships.ts`, `packages/dom/src/lint.ts`, and `docs/canonical/interaction-and-relationship-model.md` §7.
- **The data-binding mechanism** used to wire records into the field: `packages/dom/src/bind-data.ts` and `packages/dom/src/apply-recipe.ts`.
- **The data-bound Evidence Field study page** (demonstration, not experiment): `apps/site/src/pages/docs/studi`

Canonical design corroboration: `docs/canonical/natural-fields.md` (Evidence Field = electromagnetic + strong; Signal Path = `charge`/`propagate`/`fieldflow`; the electromagnetic rule), `docs/canonical/interaction-and-relationship-model.md` §26 (AI interface use cases) and §7 (`RelationshipAgent`), and `docs/canonical/authoring-and-recipes.md` §5 (the pattern schema).

1.12 Appendix B. Conversion notes (markdown → preprint)

Notation is kept LaTeX-friendly (the d, Brier, and ECE definitions and the coherence/entropy formulas translate directly to inline and display math). Figures referenced in prose but not yet drawn — the evidence-graph schematic (claims, sources, `supports`/`contradicts` edges; §3), the field/metric write-back diagram (§3.4), the three study conditions side by side (§5.2), and a sample reliability diagram with the d/criterion decomposition annotated (§5.4) — are produced at conversion time. External citations are resolved and verified against `references.md` before submission; none is invented.

1.13 Citation coverage

External citation keys/topics referenced above and resolved into `references.md` under the trust/evidence group:

- Trust calibration — trust calibration in human–automation interaction; over-trust vs. under-trust; calibrated reliance (e.g. Lee & See, *trust in automation*).
- Automation bias — automation bias / over-reliance on automated decision aids, esp. under time pressure.
- Over-reliance on LLMs — over-reliance on LLM assistants; whether confidence/explanation signals raise or reduce unwarranted reliance.
- Uncertainty visualization — uncertainty visualization; how encoding affects perceived and acted-on uncertainty.
- Source credibility — source credibility assessment and citation-inspection behavior (do readers open citations?).
- Citation faithfulness — citation faithfulness / attribution evaluation in retrieval-augmented generation (cited source not entailing the claim); LLM hallucination.
- Explanation usefulness — explanation usefulness / explainable-AI evaluation; illusory understanding; plausible rationales raising trust without improving decisions.
- Signal detection — signal-detection theory (d, criterion c) as applied to discrimination in decision tasks.
- Calibration metrics — Brier score and expected calibration error (ECE) / reliability diagrams.
- Workload instruments — a validated subjective-workload instrument (e.g. NASA-TLX).